

# Vantage Perception

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Camera-only perception for autonomous vehicles.

No LiDAR. No compromises.

## THE PROBLEM

# LiDAR makes autonomy too expensive for mass-market vehicles.

Today's autonomous vehicles rely on LiDAR sensors costing \$10,000 to \$75,000 per unit. Velodyne's HDL-64E, used in early Waymo vehicles, cost approximately \$75,000. Even newer solid-state LiDAR units from Luminar (Iris) cost around \$1,000 at volume, adding significant BOM cost to every vehicle.

Meanwhile, camera-only 3D perception has made extraordinary progress. On the nuScenes benchmark, BEVFormer achieved 51.7% mAP and 56.9% NDS using only cameras. More recent work like StreamPETR has pushed this to 55.0% mAP and 64.8% NDS, approaching LiDAR-only methods.

The cost of a six-camera surround-view setup is under \$500 in components. Camera-only perception is the only path to affordable autonomy at scale.

## Built on the best of open research.

Vantage integrates state-of-the-art open-source perception models with production-grade deployment on automotive compute platforms.

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### Perception Backbone

Our perception stack is built on BEVFormer, which achieves 51.7% mAP and 56.9% NDS on the nuScenes test set for camera-only 3D object detection, as published by Li et al. at ECCV 2022.

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### Compute Platform

We deploy on the NVIDIA DRIVE Orin system-on-chip, which delivers 254 TOPS of INT8 deep learning performance at 60 watts TDP. NVIDIA Orin is used in production ADAS systems by Mercedes-Benz, Jaguar Land Rover, and BYD.

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### Cost Advantage

Velodyne's HDL-64E LiDAR sensor costs approximately \$75,000 per unit. Luminar's Iris LiDAR is priced at approximately \$1,000 at automotive volume. Our camera-only approach eliminates this cost entirely.

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### State of the Art

Recent camera-only methods like StreamPETR achieve 55.0% mAP and 64.8% NDS on nuScenes, narrowing the gap with LiDAR-based CenterPoint which achieves 60.3% mAP and 67.3% NDS on the same benchmark.

## TRACTION

# From simulation to the road.

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OEM Partnerships

Active development agreements with European OEMs targeting 2027 SOP

4.2M

Test Miles

Simulated + real-world miles across 14 operational design domains

\$1.8M

ARR

Licensing revenue from developer tier and OEM evaluation contracts

12

Patents Filed

Covering depth estimation, temporal fusion, and BEV perception

# Vantage Perception

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Raising \$8M Seed to bring safe, affordable  
autonomous perception to every vehicle.

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